Arunbh Yashaswi

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SUMMARY

Data Scientist with 3+ years of experience at Optum, UnitedHealth Group, applying statistical modeling and machine learning across healthcare, finance, and applied NLP. Brings a systematic approach to problem solving, while mentoring junior scientists and partnering with teams to translate business needs into deployable models. Known for deep technical curiosity paired with practical execution, exploring new domains and methods to deliver data science work that is reliable and recallable to real-world impact.

Core Competencies:
Statistical Modeling | Experimental Design | A/B Testing | Machine Learning | Feature Engineering | Model Evaluation | Causal Inference | Data Analysis at Scale | Predictive Modeling | Time Series Analysis | Natural Language Processing | Recommendation Systems | Explainable Machine Learning | Risk & Decision Modeling | Data Quality | Business-Oriented Data Science

EDUCATION

Master of Science, Data Science | University of Maryland – College Park, MD | Expected in 05/2026 | GPA: 3.9/4.0
Bachelor of Technology, Computer Science Engineering | Vellore Institute of Technology – Chennai, India | GPA: 3.69/4.0

EXPERIENCE

VITG — Ellicott City, MD
Data Science Intern — 05/2025 – 08/2025

- Designed and implemented data-driven incident response and reliability workflows, applying log analysis and AI-assisted diagnostics to identify failure patterns, root causes, and remediation paths across large-scale cloud infrastructure.
- Developed automated monitoring and patching pipelines using AWS EventBridge, Lambda, and SSM, enabling systematic tracking of system state, patch outcomes, and failure signals to support consistent operational decision-making.
- Built self-service remediation and decision-support tools integrated with Slack, allowing engineers to execute approved AWS actions (EC2 restarts, Lambda triggers) directly from incident context, reducing MTTR and on-call response times by 40%.
- Analyzed operational data across $50M+ cloud infrastructure managing 200+ EC2 instances, identifying high-friction manual processes and reducing repetitive operational effort by ~35% through targeted automation.
- Designed reliability dashboards & automated reporting pipelines to surface infrastructure health metrics and incident trends, improving issue detection speed by 25% & supporting faster, data-informed collaboration across engineering & operations teams.

Optum, UnitedHealth Group — Noida, India
Data Scientist — 03/2023 – 06/2024

- Led design of large-scale NLP and document intelligence models on Azure ML, building entity extraction and layout pipelines processing 100K+ documents monthly with 50% faster processing and 18% accuracy improvement on financial documents.
- Optimized document models using noisy data, fine-tuning MPNet (84% accuracy, 14% latency reduction) and building transformer-based layout models achieving 83% zone classification and 92% table extraction on DocLayNet.
- Developed and deployed scalable inference services supporting 500+ requests per minute, reducing manual processing effort by 55% and improving deployment latency by 18% through optimized model packaging and GPU resource allocation.
- Modernized legacy ML pipelines processing 45K+ forms monthly, reducing compute time by 40%, cloud costs by $15K/month, and automating 70% of modernization tasks with GPT-4.
- Partnered with platform teams and mentored junior data scientists, translating business requirements into reproducible ML pipelines with automated retraining and monitoring, and improving developer onboarding efficiency by 40%.

Associate Data Scientist — 06/2021 – 03/2023

- Designed and productionized document intelligence systems for claims automation, engineering image-registration parser achieving 96% accuracy across 500+ form types and reducing false negatives by 35%.
- Scaled research prototypes into reliable ML services, processing 10K+ documents daily with 99.9% uptime, improving inference speed by 42% and reducing deployment cycles from 2 weeks to 3 days through reproducible CI/CD pipelines.
- Optimized OCR and layout pipelines, improving accuracy by 28%, reducing processing time by 57%, increasing throughput from 60 to 140 documents/min, and improving structure recognition from 68% to 87%.
- Established early MLOps and validation practices, building drift detection dashboards for 8 production models, reducing incident response from 48 hours to 2 hours, preventing 15+ production issues, cutting infrastructure costs by 40%, and applying causal analysis to validate model upgrades that reduced claim cycle delays by 30%.

PROJECTS

T.R.U.S.T. – Credit Risk Framework

- Built large-scale credit risk modeling pipeline by unifying Home Credit and Lending Club data (30M+ rows) and engineering 600+ temporal and vintage features to capture borrower behavior and risk dynamics.
- Trained and evaluated stacked models (Logistic Regression, XGBoost) tuned with Optuna, improving F1 score by 7 percentage points, and delivered SHAP-based reason codes to support explainability and regulatory review.

Bitcoin Sentiment Pulse – Time Series Forecasting

- Designed sentiment-augmented forecasting framework combining real-time market data with FinBERT-based news sentiment to model short-term cryptocurrency price movements.
- Evaluated multiple time-series approaches (LSTM, Prophet, ARIMA) with backtesting, showing improved directional accuracy from 61% to 67% when incorporating sentiment as exogenous signal.

SafeSim – Medical Text Simplification System

- Developed neuro-symbolic NLP pipeline combining large language models with rule-based verification to simplify medical text.
- Designed validation logic for factual consistency, achieving 95%+ accuracy in retaining key medical entities and relationships.

ACHIEVEMENTS AND PUBLICATIONS

Form Processing Using Image Matching

- Invented image similarity–based form processing method adopted across 40+ internal business applications at UnitedHealth Group, improving document matching and automation accuracy in large-scale claims workflows.

Smart India Hackathon – 1st Prize Winner (2019)

- Awarded first place in national innovation competition organized by Ministry of Education, Government of India, for developing automated customer grievance and ticket creation system adopted as proof-of-concept for large-scale public service workflows.

Low-Cost Recommendation System for Online Users

- Designed low-cost hybrid recommendation system achieving 89% accuracy in user preference prediction; published in *International Journal of Advanced Computer Science and Applications* (2020).

ADDITIONAL

Technical Skills: Python | SQL | Statistical Modeling | Machine Learning | Feature Engineering | Model Evaluation | Experimentation & A/B Testing | Causal Inference | XGBoost | Scikit-learn | SHAP | Time Series Forecasting (ARIMA, Prophet, LSTM) | Natural Language Processing | PySpark | Databricks | PostgreSQL | MLflow | ETL Pipelines | Tableau | Looker) | AWS | Azure ML
Languages: English – Native/Bilingual | Hindi – Native/Bilingual
Interests: Passionate about building data-driven AI systems that translate research into real-world use, with interests spanning applied NLP, forecasting and rapid prototyping. Enjoys staying current with ML research and exploring ideas through projects and music.